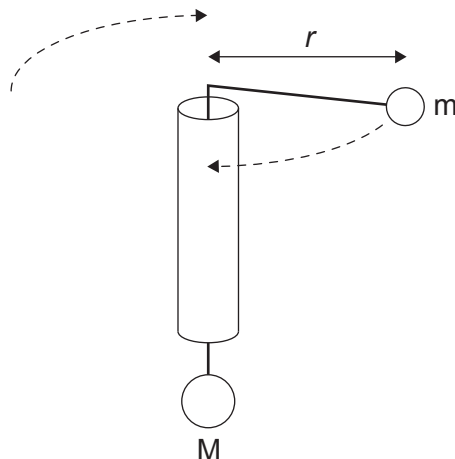


PAPER 1 / QUESTION 82

3 November 2022 / TZ0 / HL Paper 1 / original question 19 / 8822-6501

82. Two masses M and m are connected by a string that runs without friction through a stationary tube. Mass m rotates at constant speed in a horizontal circle of radius 0.25 m . The weight of M provides the centripetal force for the motion of m . The time period for the rotation of m is 0.50 s .



What is $\frac{\text{mass of } M}{\text{mass of } m}$?

- A. 1
- B. 2
- C. 4
- D. 8